

Airline Delays Analysis Design Document

1. Introduction

Dataset Description

The dataset selected for this analysis is the *Airline Delays Dataset* from Kaggle. It contains approximately 3,000 records detailing flight delays, with attributes such as:

- Airline
- Origin and destination airports
- Scheduled and actual departure times
- Date and time
- Delay durations
- Possible causes of delays

This dataset provides valuable insights into flight punctuality and factors influencing airline delays.

Hypothesis and Questions

This analysis aims to uncover trends and patterns in airline delays using Amazon Athena for querying and Amazon QuickSight for visualization. The key areas of exploration include:

1. **Airline Performance:** Which airlines experience the most frequent and longest delays?
2. **Time-based Trends:** How do delays vary by season, month, day of the week, and time of day?
3. **Airport Impact:** Which airports have the highest average delay times, and are certain routes more prone to delays?
4. **Delay Causes:** What are the most common causes of delays, and how do they differ by airline and airport?
5. **Weather Influence:** How does weather impact flight delays, and can it be correlated with delay frequency?
6. **Forecasting Trends:** Is there a trend in delay duration over time, and can we forecast future delays?

2. Services Used and Rationale

Amazon Athena

- **Purpose:** SQL-based querying and data analysis.
- **Rationale:**

- Serverless architecture enables efficient querying without infrastructure management.
- Supports direct analysis of structured data in S3.
- Cost-effective for large queries

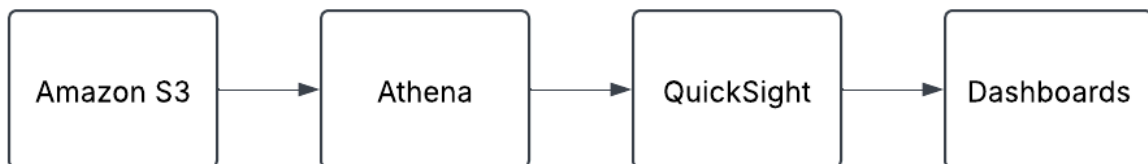
Amazon QuickSight

- **Purpose:** Data visualization and dashboard creation.
- **Rationale:**
 - Enables interactive and shareable visual reports.
 - Provides machine learning-powered insights for trend analysis.
 - Integrates well with Athena for seamless data visualization.

3. System Diagram

Architecture Overview

1. **Data Storage:** The dataset is stored in **Amazon S3**.
2. **Querying & Processing:** Amazon Athena is used for structured SQL analysis.
3. **Visualization:** Insights are visualized using Amazon QuickSight dashboards.
4. **Additional Enhancements** (if needed):
 - AWS Glue for ETL (if transformation is required).



4. Analysis Plan

Data Processing

- Clean data by handling missing or inconsistent values.
- Normalize timestamps for consistent time-based analysis.

Querying Strategy

- Aggregate delays by airline, airport, and time factors.
- Identify relationships between weather conditions and delay frequencies.
- Compute average delay durations for different routes and carriers.

Visualization Approach

- **Bar Charts:** Airline performance comparison.
- **Heatmaps:** Seasonal and daily delay trends.
- **Line Graphs:** Forecasting delay trends over time.
- **Geo Maps:** Airport-specific delay distributions.

5. Conclusion

This project aims to provide actionable insights into airline delays, helping improve operational efficiency in the aviation sector. Using Amazon Athena for analysis and Amazon QuickSight for visualization ensures an efficient, scalable, and insightful approach to identifying and mitigating flight delays.